

Instructor: Date:

Learner: Regional Branch:

Program: Center:

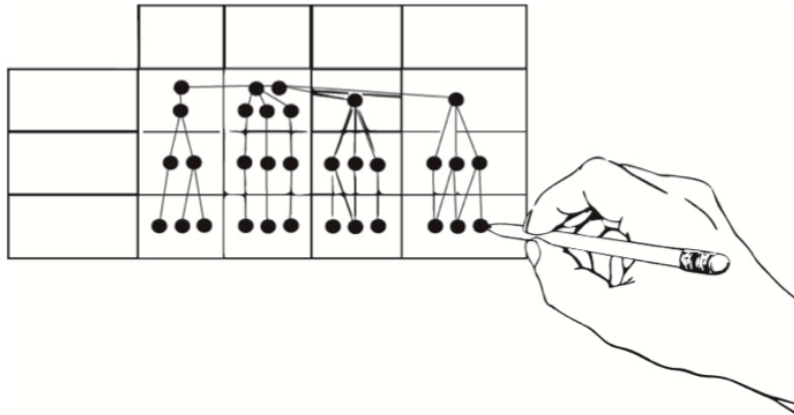
CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Let's learn about software design patterns!



Learning Outcome:

Identify key information about people's jobs and places of work in spoken or written contexts.
Make the information system design report, based on the selection of both hardware and software tools required for the IT solution.

Can Do Statements:

- I can talk about commonly occurring problems in software design.
- I can explain what design patterns are.
- I can ask questions related to design patterns in software engineering.

Ask your partner:

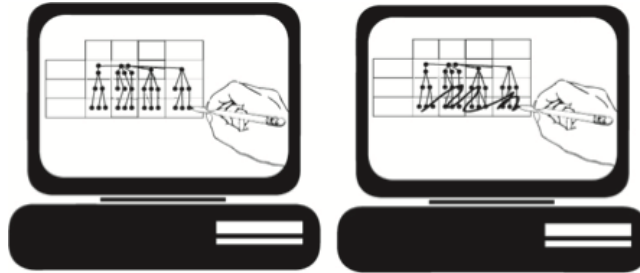
- What are the most common errors you find in software design?
- How do you solve common errors in software design?





Let's Explore!

We have got some software design problems!



Instructions: Read the text below.

Mr. Dawson: Good morning, Helen!

Helen: Good morning, Mr. Dawson!

Mr. Dawson: Can you please come to my office?

Helen: Yes, Sir.

Mr. Dawson: I have this report about some problems that occur with our software design, is this yours?

Helen: Yes, sir. It's mine.

Mr. Dawson: So, I'm all ears.

Helen: Indeed, we have got some software design problems. In our last survey of customer support system, there were fewer positive results than two months ago. Users complain that they don't receive an immediate response to their requests. This is due to some weaknesses in the software system. On the other hand, we need to create a chat application so they can communicate with each other.

Mr. Dawson: What do you suggest?

Helen: We must implement design patterns, which are very important tools in software design. They can be descriptions or templates that solve the most common problems in software design. Such solutions must be reusable, or at least need to have a variant that can figure out other problems as subtle issues. Because if these ones aren't solved on time, it could be worse than solving the big ones afterward. For that reason, design patterns could speed up and improve our software development process.

Mr. Dawson: You're right. It's better to implement this solution than other complex ones.

Helen: Exactly.

Mr. Dawson: What other advantages do design patterns offer?

Helen: Coders and architects have got the possibility to improve their code readability.

Mr. Dawson: How does it work?

Helen: It works on different sections, such as participants, collaboration, and structures.

Mr. Dawson: Have our engineers got the appropriate experience on that? Or, do we have to hire engineers with more experience than theirs?

Helen: Of course, as software engineers, they have got a lot of training in this practice.

**Let's Practice!**

Helen is talking to the engineers about the software problems.

Instructions: Use the words from the Word Bank to complete each sentence.

Word Bank

got / have / have got / has got

1. You _____ to improve the software so that it has an immediate answer to requests.
2. We _____ got to create a chat application, users are complaining because we don't have one.
3. One of the engineering teams _____ to deal with the bugs in the software immediately.
4. Another team has _____ to check all the software weaknesses reported by the users.

**Let's Practice!**

Helen is explaining why design patterns are the best solution.

Instructions: Read the sentences. Write the missing word(s) in the blank. Use: less, worse, more, better.

1. Design patterns are a _____ programming solution than most.
2. Design patterns are _____ useful than other solutions, since they are reusable.
3. Other design solution ideas are _____ effective at eliminating bugs than design patterns.
4. It is better to solve small problems with design patterns than letting them become _____ and impossible to solve.



Extension Activity

Planning my Design Pattern.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about possible design problems.
- Plan the way you would solve that problem using a design pattern.
- Make a brochure in which you show your plan to solve the problem.
- Share your brochure with the rest of the rest of the class.

Self-Assessment

Yes No

- I think about possible software design problems.
- I plan a design pattern to solve the problem.
- I present my plan to a partner and receive suggestions.

☐	☐
☐	☐
☐	☐



Target Words

- **speed up** (phrasal verb): to perform a task quickly
e.g. Design patterns could speed up and improve development software processes.
- **request** (noun): a petition done by a user
e.g. Users complain that they don't receive an immediate response to their requests.
- **coder** (noun): a programmer in the computing field
e.g. Coders and architects have got the possibility to improve their code readability.
- **pattern** (noun): a scheme to follow for solving a coding problem
e.g. We must implement design patterns.
- **training** (noun): the development of particular skills before making an activity
e.g. As software engineers, they have got a lot of training in this practice.
- **subtle** (adjective): imperceptible
e.g. This solution must be reusable or at least to have a variant that can figure out other problems as subtle issues.



Learning Points

Comparatives With Irregular Adjectives.

You use **comparatives** to show differences between two or more elements.

In English, there are regular and irregular adjectives. Regarding regular adjectives, there are clear rules to establish their comparison, such as "er" or "ier".

On the other hand, irregular adjectives don't follow exact rules since they have different changes.

Subject + verb + **irregular adjective** + **than** + complement

Examples:

*This design pattern is **better than** that other one.*

*The performance of this application is **worse than** the previous ones.*

IRREGULAR ADJECTIVES	
Good / Well	Better
Bad	Worse
Far	Farther / Further
Little	Less
Many / Much	More
Old	Older / elder

Examples

- In our last survey of customer support system, there were fewer positive results than two months ago.*
- If these ones aren't solved on time, it could be worse than solving the big ones afterward.*
- It's better to implement this solution than other complex ones.*





Learning Points

Possessive Pronouns.

You use **Possessive Pronouns** to indicate the possession of something. In English, for each subject pronoun there's a *Possessive Pronoun*.

These are the **possessive pronouns**: *mine, yours, his, hers, its, ours, and theirs*.

Check the chart below with examples for each pronoun.

POSSESSIVE PRONOUNS		
Subject Pronoun	Possessive Pronoun	Example
I	Mine	- Is this your design? - Yes, It's mine.
You	Yours	I found this report in my desk, is it yours?
He	His	- I bet this computer belongs to John. - No, it doesn't. This one is blue and his is red.
She	Hers	- Whose report is this? - Let me check, maybe it's my assistant. Oh...yes, it's hers.
We	Ours	Hey, the software is ours. Don't touch it.
They	Theirs	- I love the design Karen and David create - I prefer Cindy and Christian's. Even though theirs is good too.

Examples

- I have got this report about some problems that occur with our software design, is this yours?*
- Yes, sir. It's mine.*

